# **Exp. No: 2** Determination and plotting of Electric field and Electric flux density in sphere full of charges with varying radius.

## Aim:

To write program for electric field and electric flux density in sphere using SCILAB.

## Software required:

•SCILAB version 6.0.1

Code:

EXP 1 EME.sci | EXp 1)ii.sci | exp 1)iii eme.sci | exp 2.sci | exp 2 case 1.sci

```
clc;
clear;

q=input('Enter the value of q=');
r=input('Enter the value of r=');

efcilon=8.854*10^-12;
pi=3.14
E=q/4*pi*efcilon*r^2;
disp("The electric field is", [E]);
D=E*efcilon;
disp("The flux density is", [D]); x = linspace(0, 10, 50);
y = E * exp(-0.1 * x);
z = D * exp(-0.1 * x);

figure();
subplot(2, 2, 1);
plot2d3(x,y);
xlabel("R");
ylabel("Electric field intensity");
title('Electric field intensity');
subplot(2, 2, 2);
plot2d3(x,z);
xlabel("A");
ylabel("Electric flux intensity");
title('Electric flux intensity');
```

output:

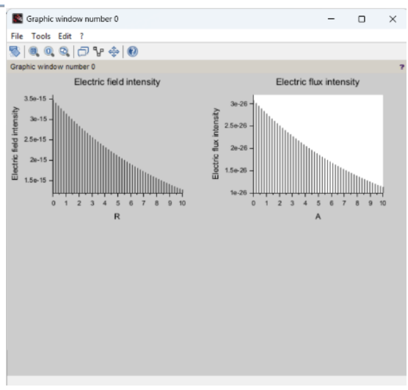
```
Startup execution:
  loading initial environment

--> exec('C:\Users\jasiy\OneDrive\Documents\exp 2.sci', -1)
Enter the value of q=5*10^-6

Enter the value of r=10

  "The electric field is"
   3.475D-15
  "The flux density is"
   3.077D-26

-->
```




## **Theoretical calculation:**

EXP No. : 2

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~~case (i)~~

Output :

Enter the value of q = $4 \times 10^{-6}$

Enter the value of r = 5 cm.

Electric flux Density (D) vs Radius graph is generated successfully.

"The electric field is" 6.950 D-16

"The flux density is" 6.154 D-27

Case (i)

Output :

Enter the value of q = $5 \times 10^{-6}$

Enter the value of r = 3 cm

Linear Density : D vs Radius graph is successfully generated.

Case (ii)

output :

Enter the value of q = $5 \times 10^{-6}$

Enter the value of r = 10 cm

Electric flux Density (D) vs Radius graph is generated successfully.

Case (iii) : output :

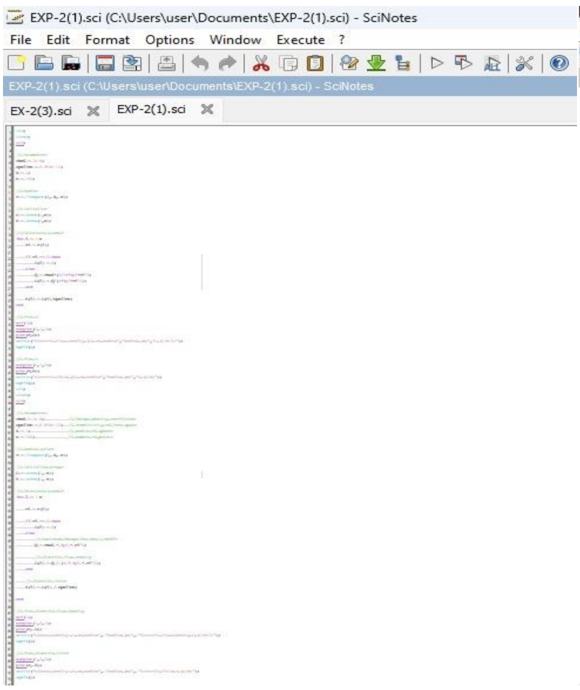
Enter the value of q = $3 \times 10^{-6}$

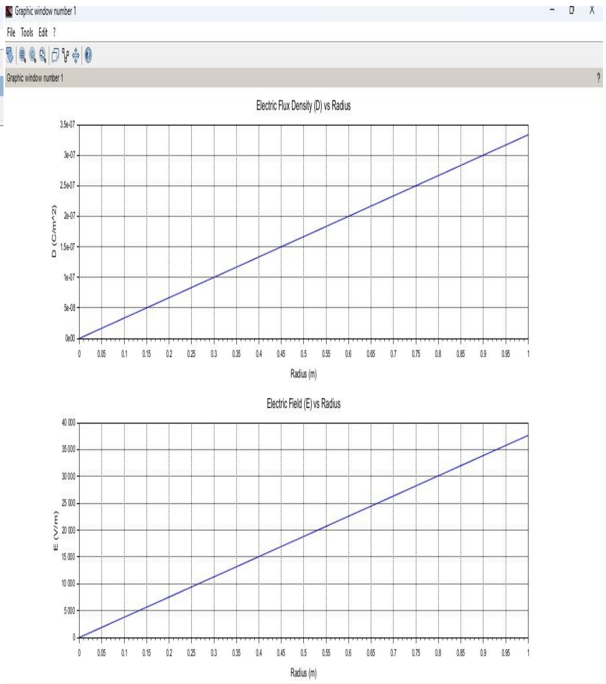
Enter the value of r = 1mm

## Case 1:

**Derive and plot the electric field (E) and electric flux density (D) inside a uniformly charged sphere with charge density ρ0. What are the values of E and D at the center and at the surface of the sphere?**

**Code:** **output:**

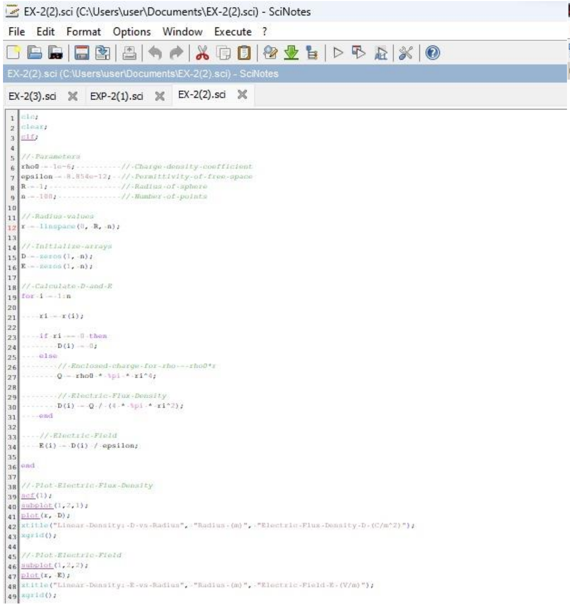


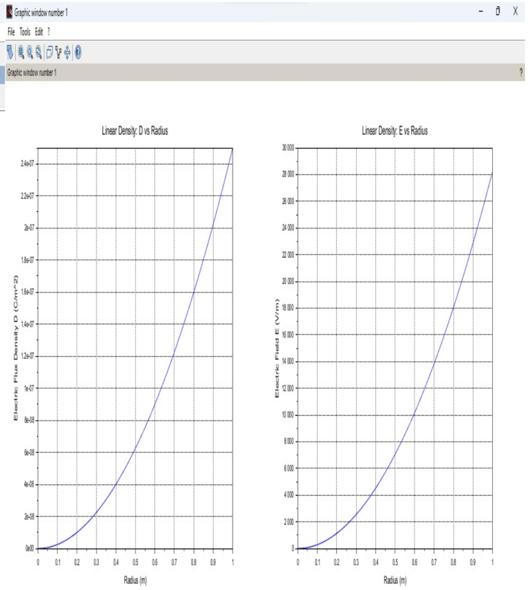




## Case 2:

**For a sphere with a linearly increasing charge density ρv=ρ0r.derive the expressions for the electric field (E) and electric flux density (D). Plot their variations inside the sphere and analyze the trends.**

**Code:** **output:**



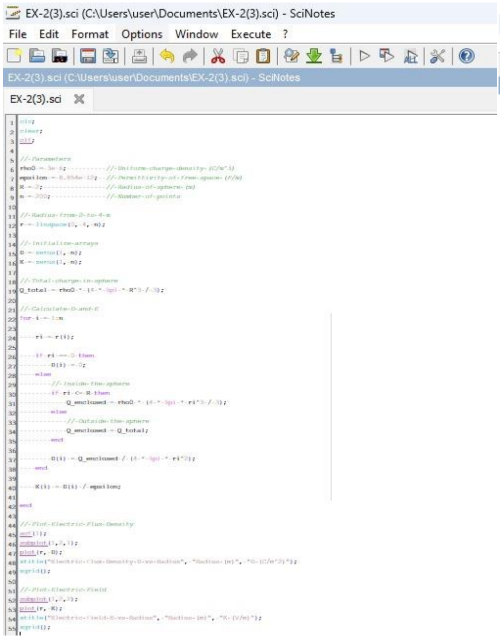




## Case 3:

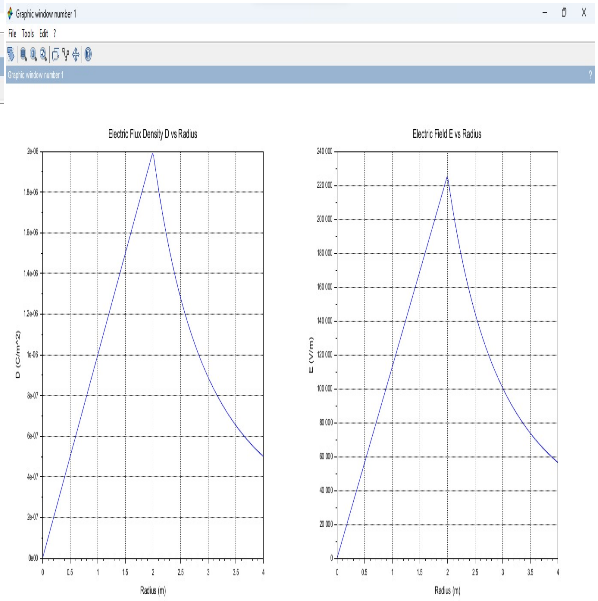
**A sphere of radius R=2 m has a uniform charge density ρ=3×10^−6 C/m^3**

1. **Calculate and plot the electric flux density D(r) and electric field E(r) as a function of radial distance r, ranging from 0 to 4 meters.**
2. **Discuss the behaviour of the electric field inside and outside the sphere.**

**Code:** **output:**



```
clc;
clear;
clf;

// Parameters
rho0 = 3e-6;          // Uniform charge density (C/m^3)
epsilon = 8.854e-12;  // Permittivity of free space (F/m)
R = 2;                // Radius of sphere (m)
n = 200;              // Number of points

// Radius from 0 to 4 m
r = linspace(0, 4, n);

// Initialize arrays
D = zeros(1, n);
E = zeros(1, n);

// Total charge in sphere
Q_total = rho0 * (4 * %pi * R^3 / 3);

// Calculate D and E
for i = 1:n

    ri = r(i);

    if ri == 0 then
        D(i) = 0;
    else
        // Inside the sphere
        if ri <= R then
            Q_enclosed = rho0 * (4 * %pi * ri^3 / 3);
        else
            // Outside the sphere
            Q_enclosed = Q_total;
        end

        D(i) = Q_enclosed / (4 * %pi * ri^2);
    end

    E(i) = D(i) / epsilon;

end

// Plot Electric Flux Density
scf(1);
subplot(1,2,1);
plot(r, D);
xtitle("Electric Flux Density D vs Radius", "Radius (m)", "D (C/m^2)");
xgrid();

// Plot Electric Field
subplot(1,2,2);
plot(r, E);
xtitle("Electric Field E vs Radius", "Radius (m)", "E (V/m)");
xgrid();
```




## Result :

Thus the program was verified for electric field and electric flux density in sphere using SCILAB was successfully.